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Document Version History

Version	Date	Authors	Description of Changes
1.0	October 10, 2026	Team CodeLex	Initial definition of the problem, system overview, and background.
2.0	October 17, 2026	Team CodeLex	Scope refinement following a client meeting and technical guidelines: console-based interface with a keyboard-navigable menu, data persistence via JSON files, report export in CSV format, a 6-tier role hierarchy managed via email by the Director, classification of active/passive cases by legal category, and detailed logging of expenses/debts per case.

Problem

The daily administration of the law firm faces a critical bottleneck due to its reliance on manual tools, such as spreadsheets, to track legal proceedings. This approach consumes an excessive amount of administrative time, leads to disorganized legal documentation, and complicates the monitoring of litigation-related expenses and collections—all of which negatively impact overall efficiency and case traceability. Because various roles—including the Principal Director, Partners, Associate Attorneys, Junior Attorneys, Legal Assistants, and the Secretary—interact with records without email-based credential restrictions, it is impossible to properly enforce role-based permissions or track which user modified a specific file, thereby compromising the integrity of legal data. Another significant drawback of this manual management system is operational fragmentation and the lack of a structured case classification framework. Currently, staff waste valuable time because they lack a search tool capable of quickly filtering active and inactive cases by unique case ID, litigation code, client name, type of proceeding, priority level, or legal practice area (such as Criminal, Family/Support, Civil, or Labor law). Furthermore, spreadsheets do not provide a structured chronological calendar or a task log where attorneys can record procedural developments, nor do they allow staff to check upcoming hearing dates or update pending tasks assigned to specific judicial units. The firm also lacks a transparent mechanism for recording litigation expenses and client balances within individual case files, as well as the ability to generate standardized reports. Without a detailed expense ledger integrated into the case file, lawyers

and administrative staff cannot easily record specific legal costs—such as court fees, notarized copies, or administrative expenses—along with their descriptions and amounts. This makes it difficult to instantly verify how much a client owes, the reason for a charge, or whether a balance has been paid in full. It is therefore urgent to implement a secure, modular system that centralizes case files, manages user roles and permissions, provides calendar access, and tracks finances—utilizing structured data persistence via JSON files and the automatic generation of reports in CSV format.

Overview

LegalSync is a legal case management system designed with an object-oriented, modular architecture, created to transform how a firm handles its daily operations—moving away from slow, error-prone manual spreadsheets. Operating via a navigable menu with keyboard-based data entry, the system serves as a digital hub that consolidates client profiles, defendant records, active and closed case files, judicial unit assignments, legal filings, hearing schedules, and detailed client debt records into a single platform. Access to the system is governed by a hierarchical, email-based role management module. At the top of the firm's hierarchy, the Principal Director holds full administrative privileges, allowing them to grant user access, review accounts, and assign or modify roles—linked to each staff member's email address—across six distinct profiles: Principal Director (DIRECTOR), Partner (PARTNER), Associate Lawyer (ASSOCIATE_LAWYER), Junior Lawyer (JUNIOR_LAWYER), Legal Assistant (LEGAL_ASSISTANT), and Secretary (SECRETARY). While lawyers can issue procedural filings, add tasks, and update pending actions, the Secretary profile is strictly limited to read-only access, covering case statuses, the calendar, and records of client expenses and billing. Beyond storing data persistently in system-generated JSON files, LegalSync optimizes legal workflows through a multi-criteria search engine that retrieves case files based on Case ID, Client Name, Active/Closed status, or Legal Category (Criminal, Family, Civil, or Labor Law). It also includes a scheduling and chronological calendar module, where staff can record and consult hearing dates and procedural tasks. Furthermore, opening any case file immediately displays the associated client profile, along with a detailed statement of expenses and debts—listing every legal cost incurred, including its description, amount, and payment status (Pending or Paid in Full)—with the option to export this information and the status of cases into CSV-format reports.

Background

In law firms, the structured management of case information is essential for maintaining proper oversight of judicial proceedings, organizing legal documentation, and providing transparent services to clients. Lawyers, paralegals, and administrative staff must continuously update case statuses, categorize proceedings by area of law and priority, coordinate procedural tasks with judicial offices, and consult records related to clients, opposing parties, and litigation expenses. As the number of active and pending cases grows, these tasks become increasingly complex, requiring a well-organized software-based

approach. Currently, the law firm manages its case and billing information using Microsoft Excel spreadsheets. While this tool allows for the recording of basic data, it has significant limitations when it comes to managing relationships between clients, categorized case files, linked documents, detailed debts, and system users. Manual updates are time-consuming, increase the risk of human error, and make it difficult to maintain consistent, up-to-date information across different roles. Another significant challenge lies in document organization, report generation, and the traceability of modifications. Documents associated with various cases are often stored separately from the case files, making them difficult to locate quickly. Furthermore, the lack of a detailed change log prevents the Principal Director from tracking actions taken within the system or identifying which user modified a specific case file or payment record. To address these issues, the development of "LegalSync" is proposed—utilizing JSON file persistence and CSV report generation—to centralize information regarding clients, categorized case files, hierarchical user accounts, calendars, detailed expense records, and associated documents.